Preliminary analysis of NGT cluster usage and proposal

Felice Pantaleo (CERN)
July 2026

Method

Data

Extracted pod lifecycle, requests, placement, cordons and per-GPU utilization from cluster monitoring, 30 days at 5 min resolution.

Scope

1447 GPU/MIG requests from 111 users on the on-premise pools. STEAM Academy accounts and cloud T4 nodes excluded.

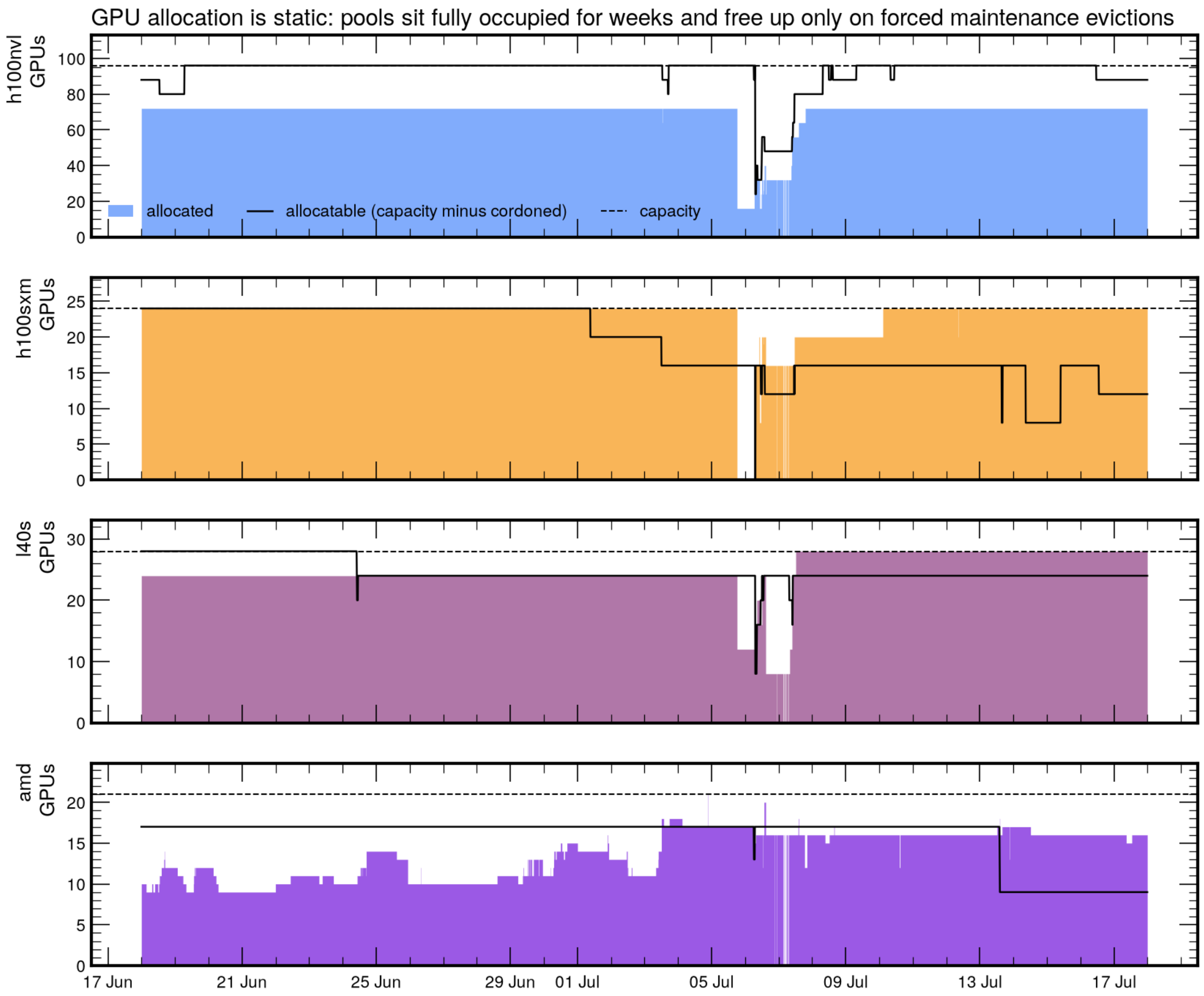
Attribution

Users mapped to WP1 to WP3 via the project roster, group and manual classification. WP1 users possibly underestimated.

> **A request = one pod asking for GPUs. Wait = creation to scheduling.**
> **Idle = GPU utilization below 5%.**

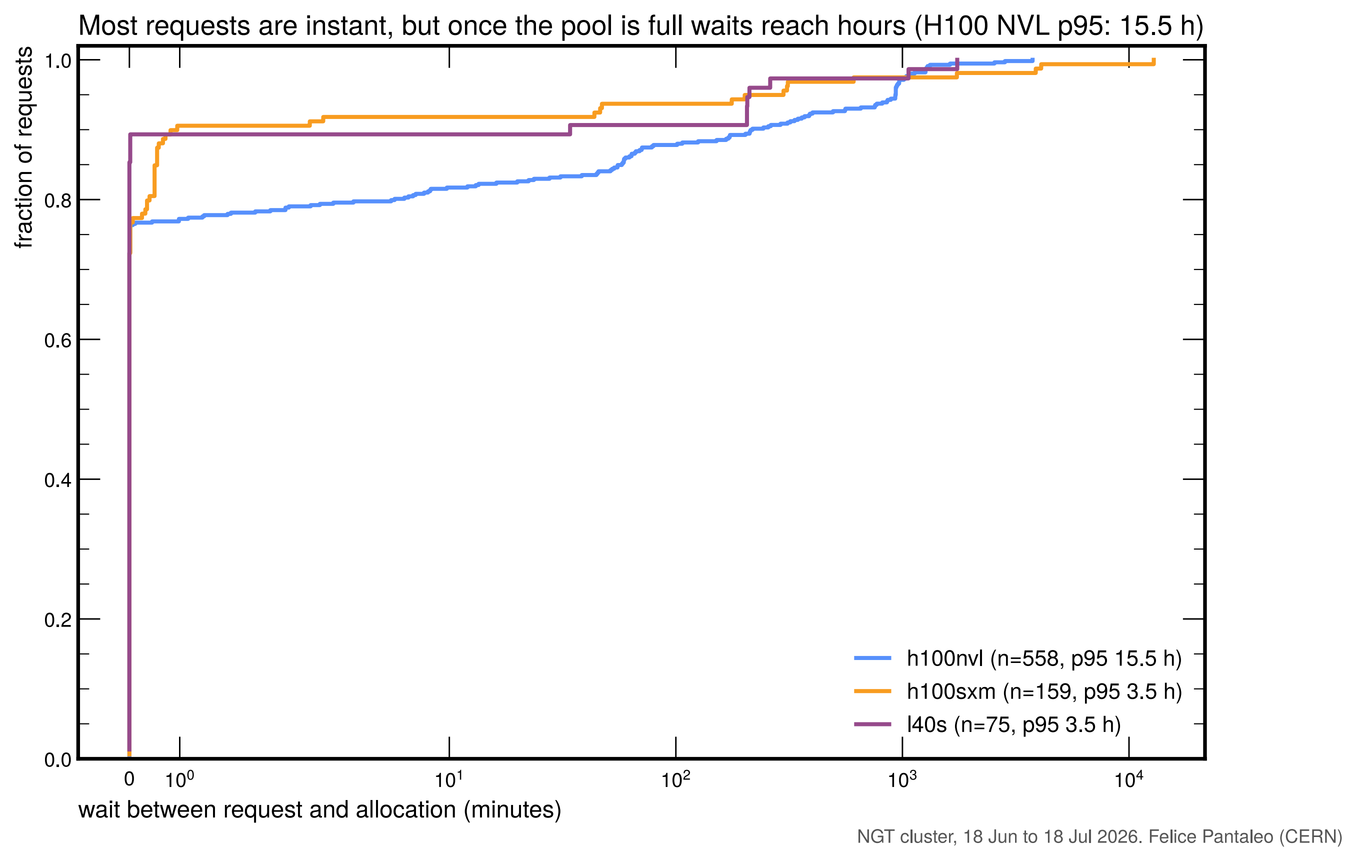
Static allocation

- All pools at their effective ceiling around the clock
- No diurnal variation: allocation follows ownership, not workload
- Only the forced maintenance evictions of 6 to 8 July freed capacity



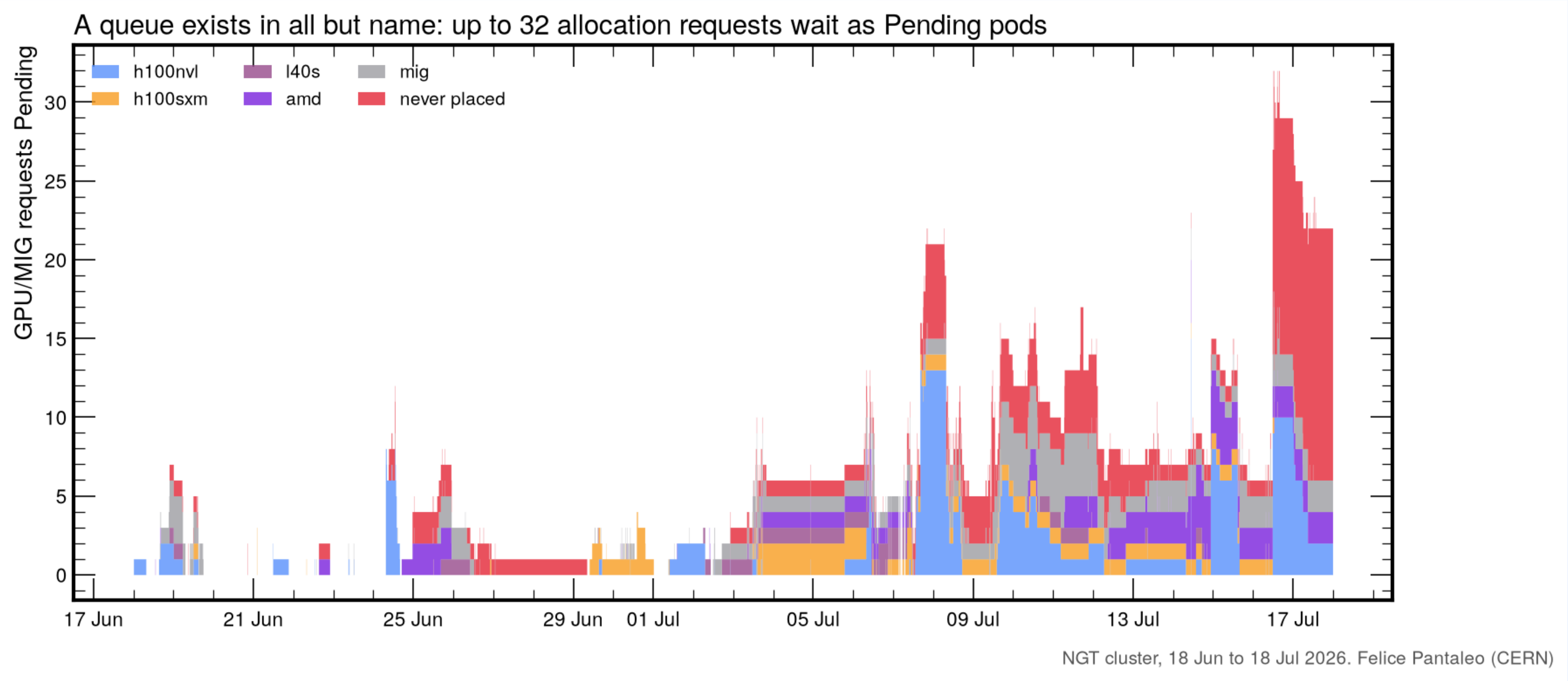
NGT cluster, 18 Jun to 18 Jul 2026. Felice Pantaleo (CERN)
NVL: MIG-enabled GPUs are not visible to DCGM, so the full-GPU ceiling sits below the 96-GPU capacity line. AMD from user-pod accounting (no DCGM).

Waiting times



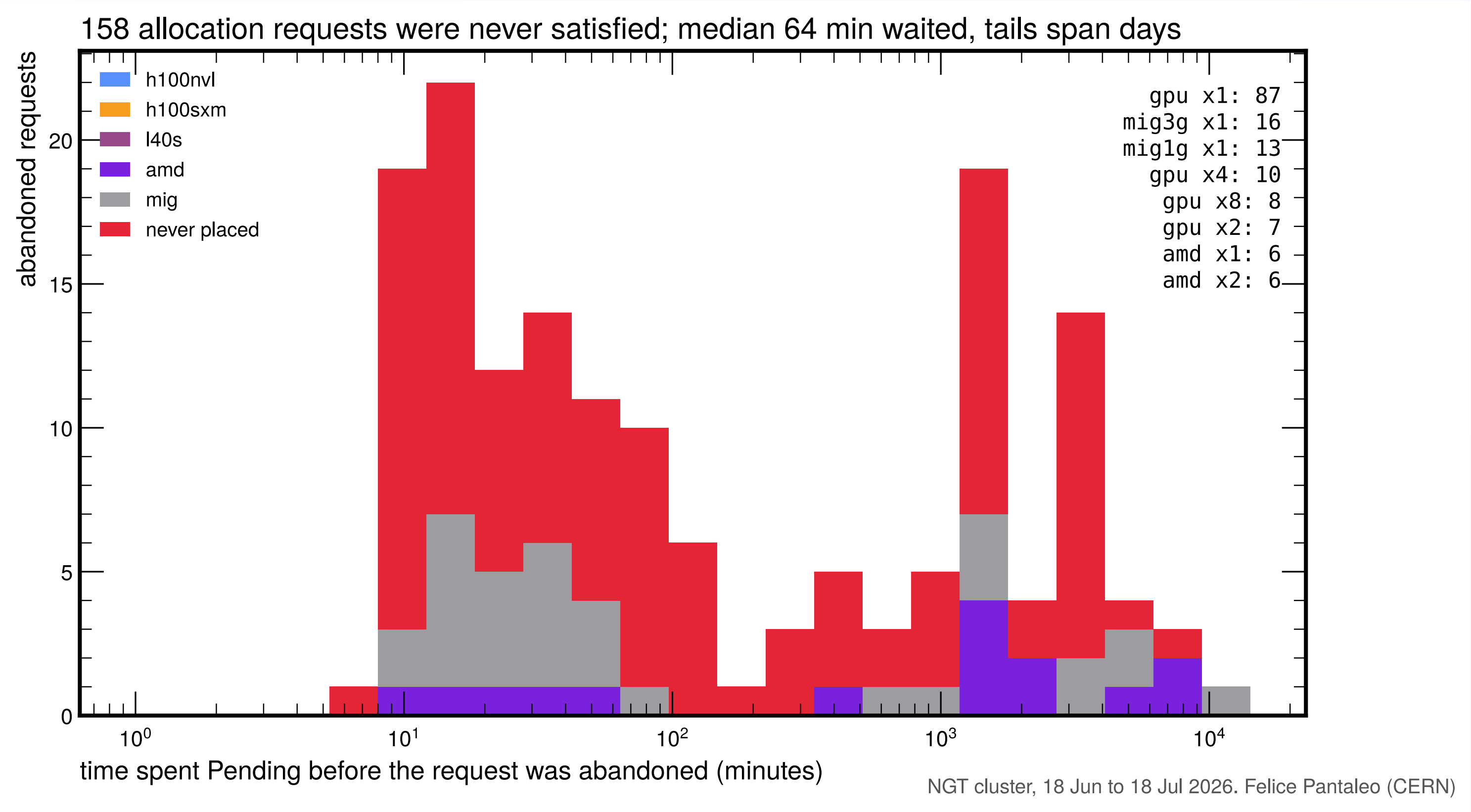
75 to 90% of requests are satisfied immediately; the H100 NVL p95 wait is 15.6 h.

The Pending queue



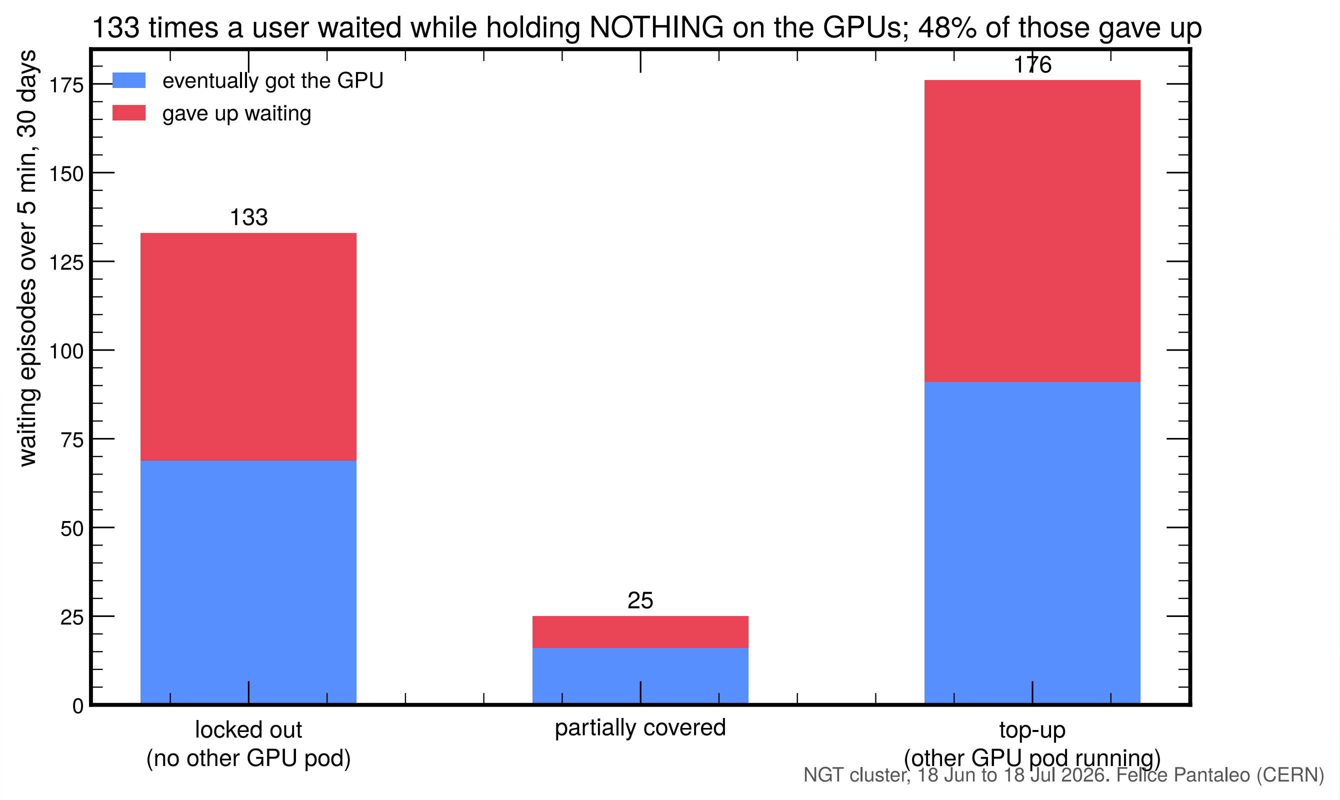
Peak backlog: 32 simultaneous Pending requests, stacked by target pool.

Unsatisfied requests



146 requests (10%) were never satisfied; their owners gave up after a median 72 minutes of waiting.

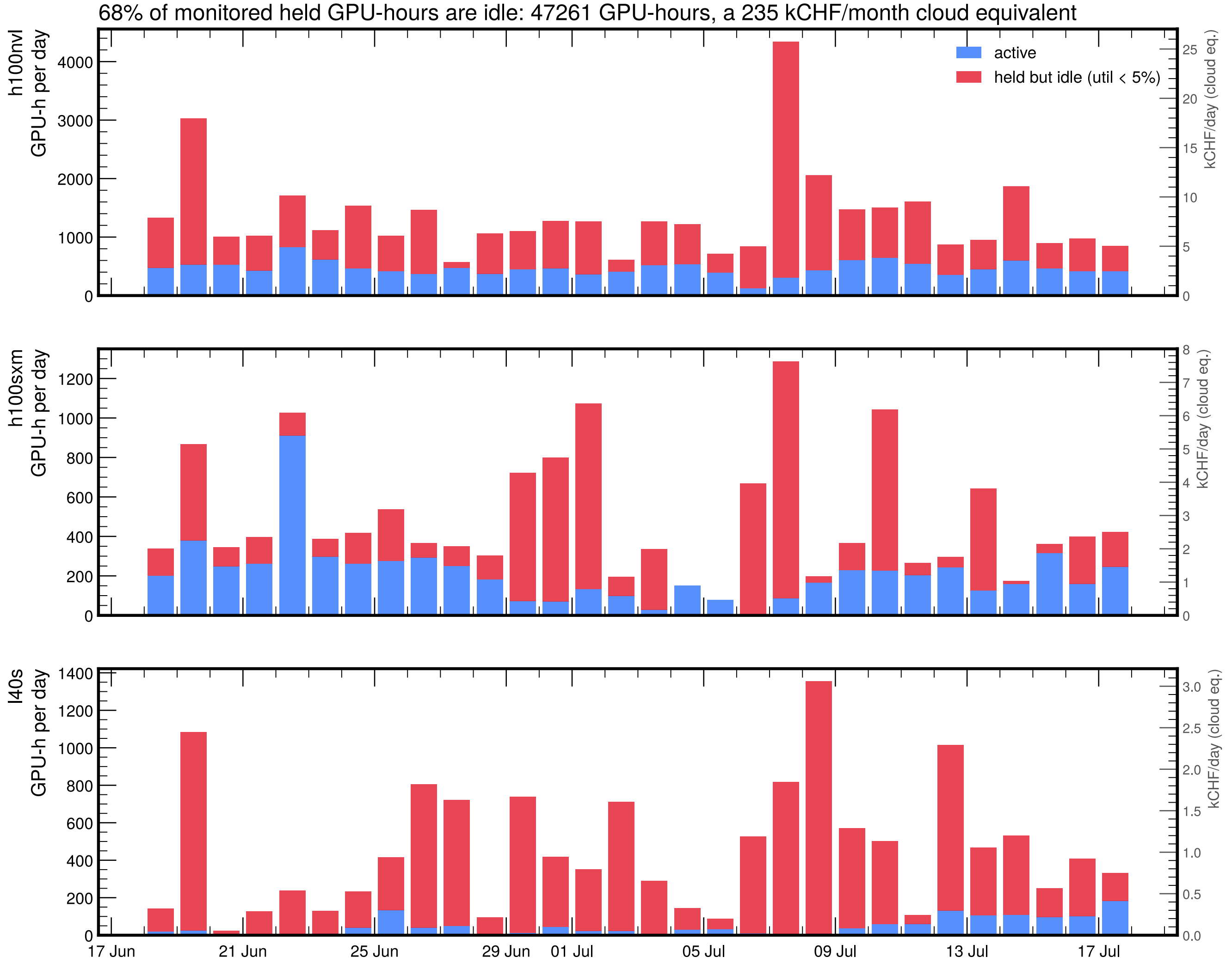
Lockouts



131 times last month a new user not has been locked out at their first request.
Over half of all queue pressure is top-up demand from users already running.

Idle held GPU-hours

- 47 000 GPU-hours parked in 30 days
- 235 kCHF/month cloud equivalent
- Right axes: kCHF/day per pool
- Continuous idle baseline on NVL and L40S; bursty parked episodes on SXM



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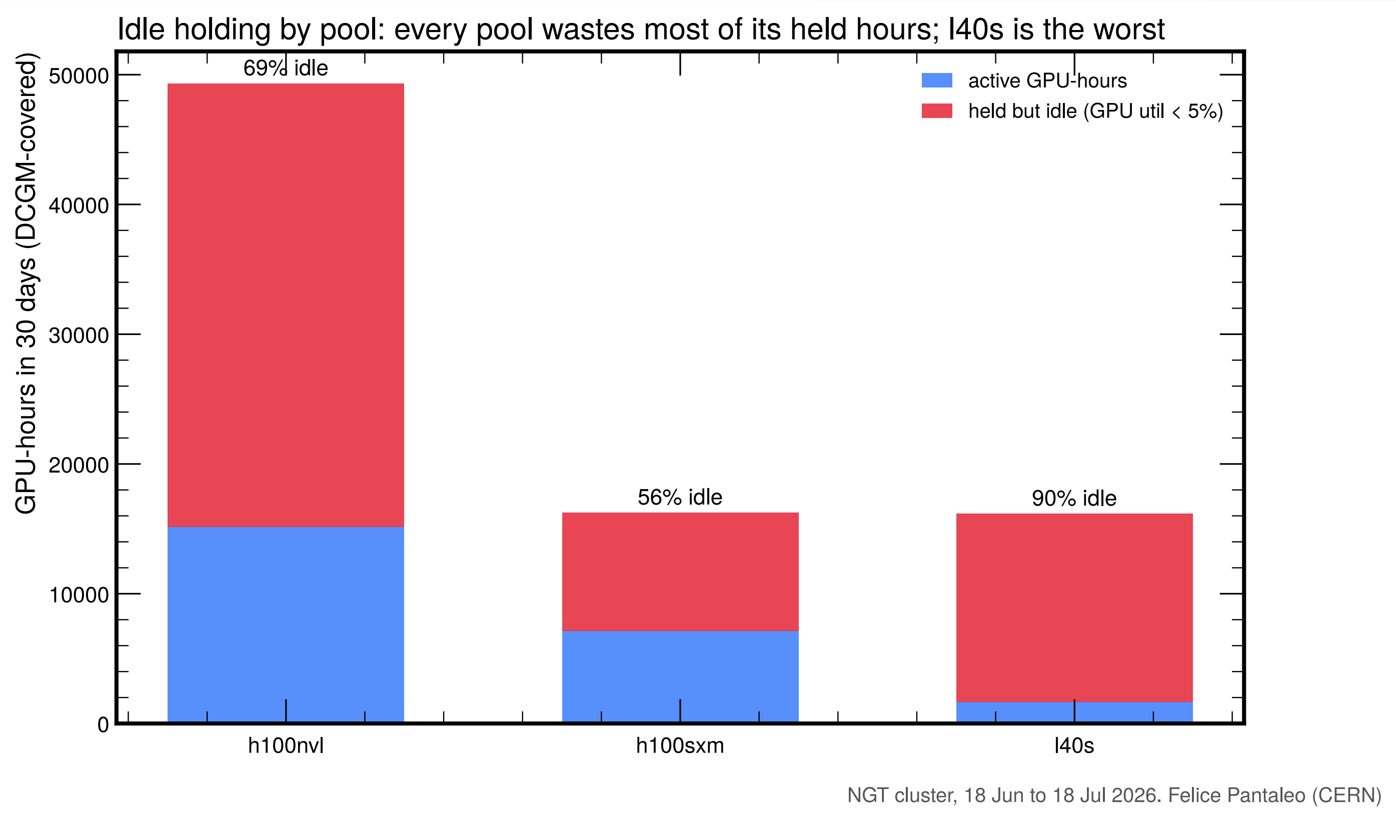
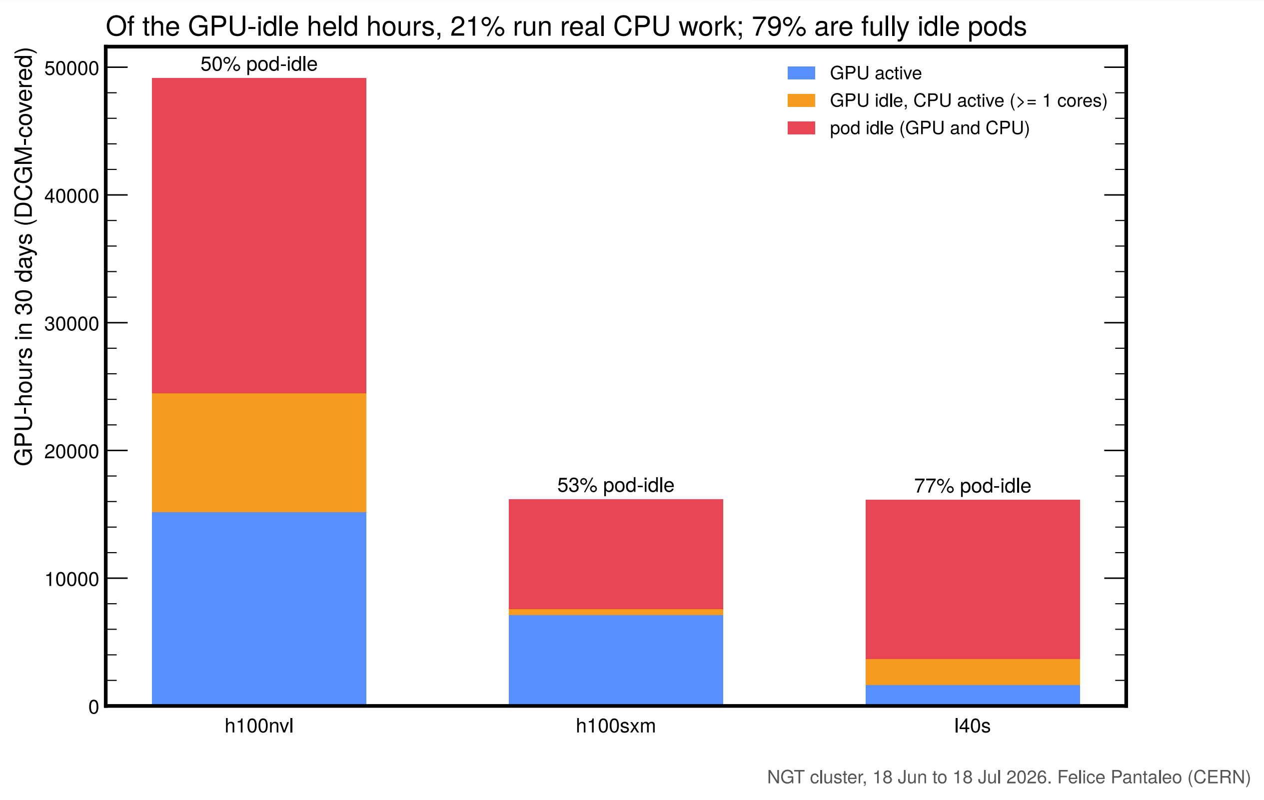
DCGM utilization available for 545 full-GPU allocations; MIG slices and AMD not covered. Cloud rates as in the cost plot.

Idle share by pool



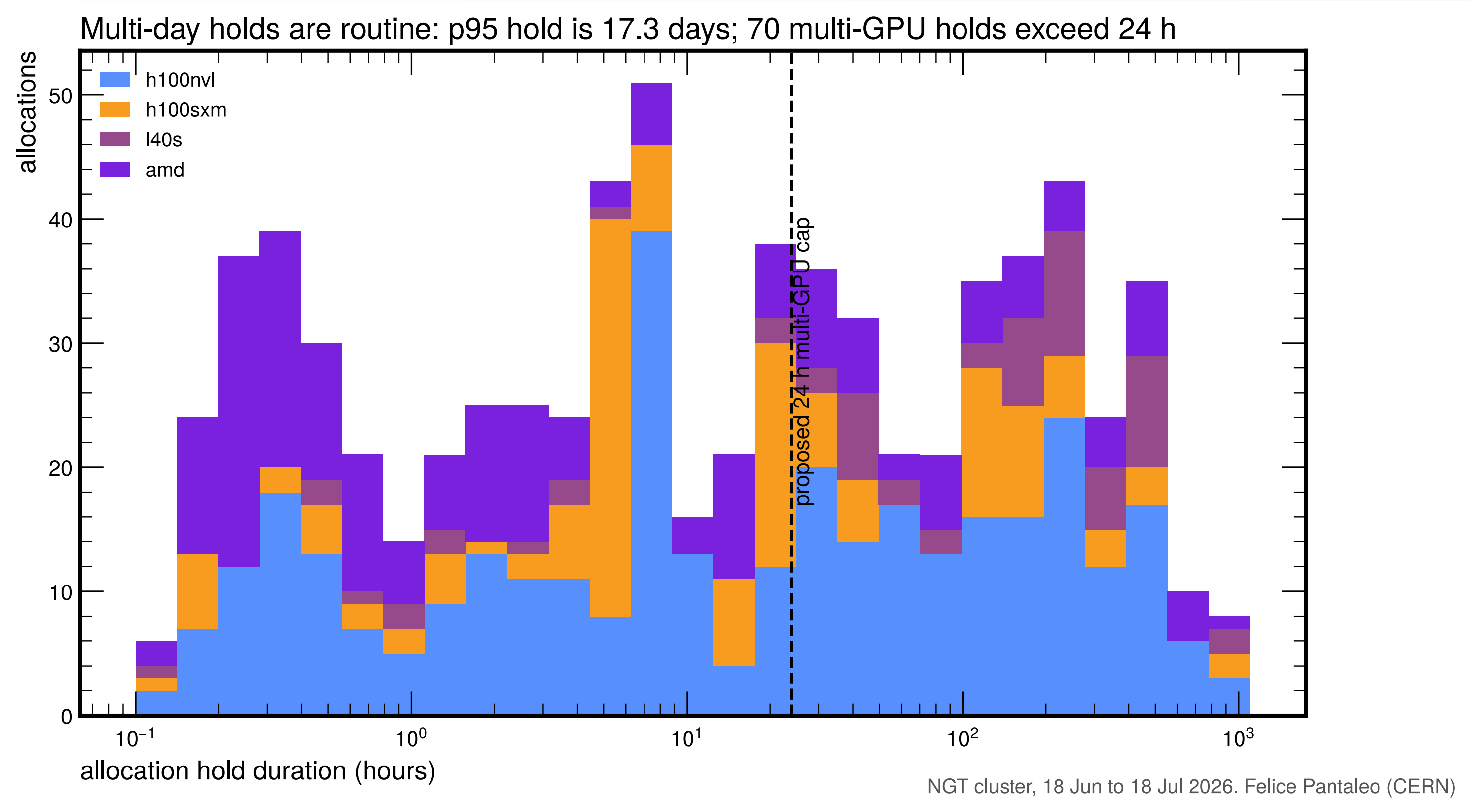
The L40S pool is idle for 91% of its held hours: mainly explained by spiky heterogeneous algorithms development, benchmarking, and fear of losing the pod/tmux session.

GPU-idle vs pod-idle



Only 21% of GPU-idle hours run real CPU work (>= 1 core); 79% are fully idle pods. Robust across 0.5 to 2 core thresholds.

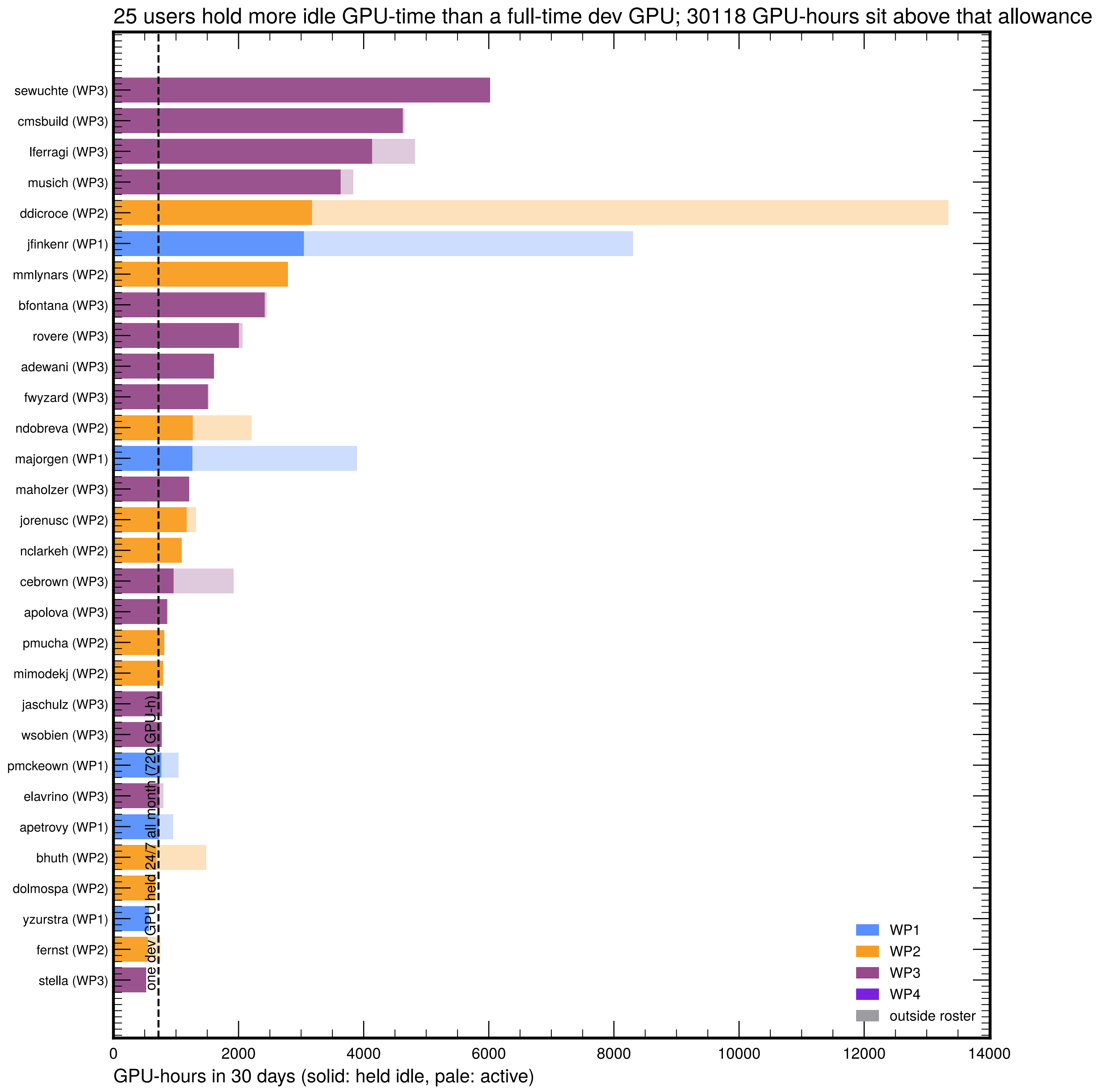
Hold durations



The p95 hold is 16.7 days; 68 multi-GPU holds exceed 24 hours.

Heavy idle holders

- Solid = held idle, pale = active
- Dashed line: one dev GPU holding 24/7 all month (720 GPU-h).
- **25 users** exceed it
- **30 100 GPU-hours** sit above it
- The consumption and idleness rankings select different users: the top
  consumer is almost entirely active



NGT PMC, July 2026

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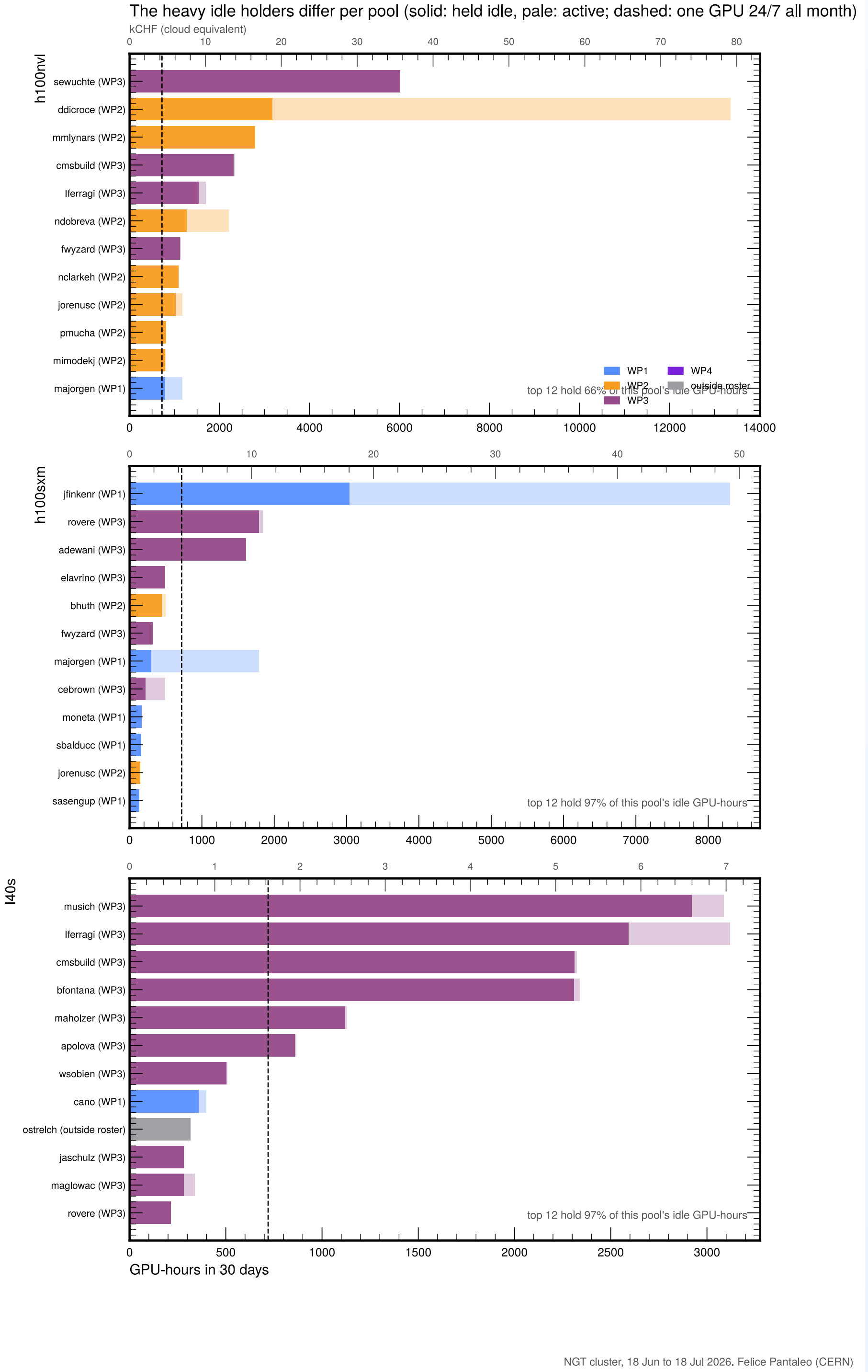
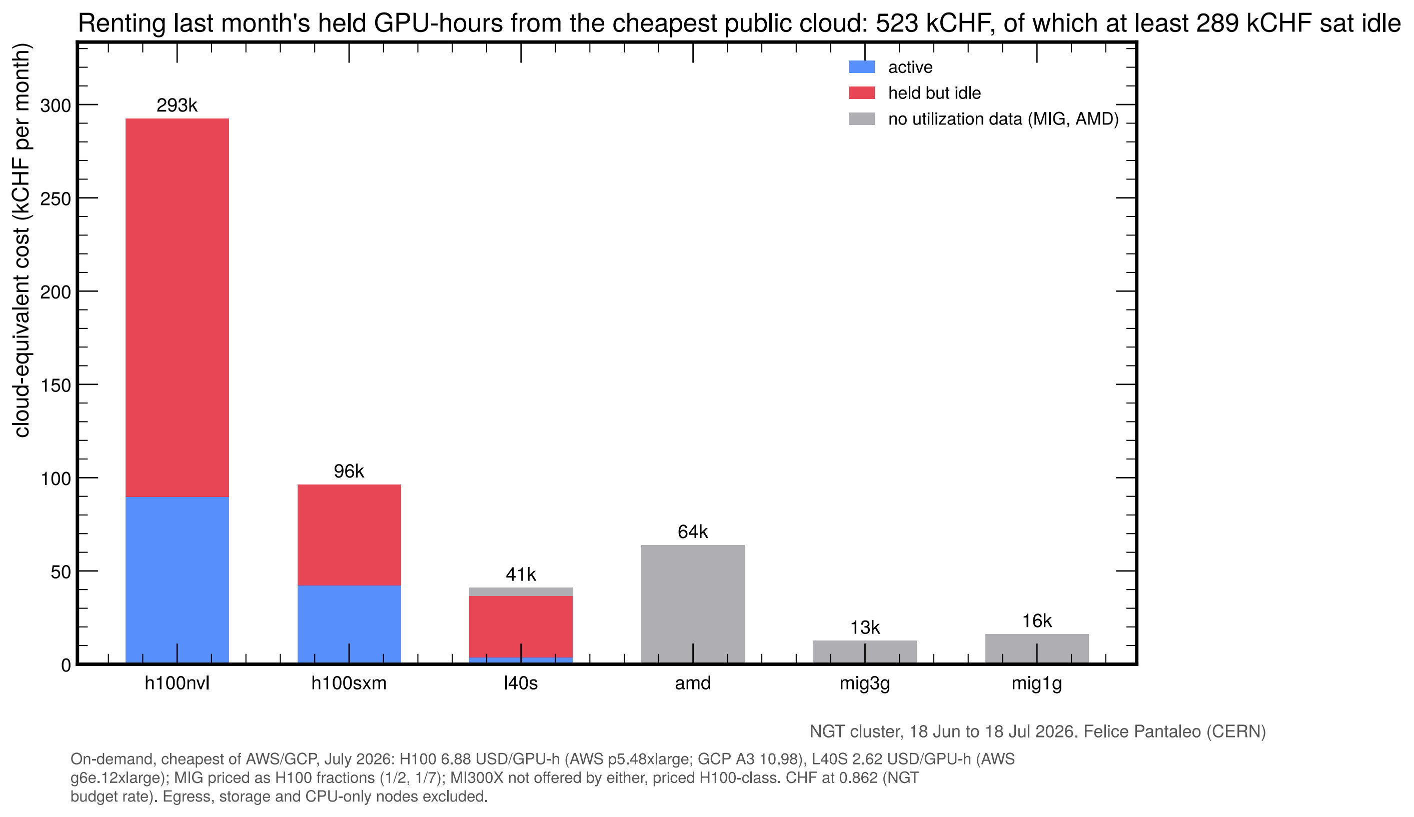
Idle holders by pool

- H100-NVL: broad behavior, top 12 hold 66% of idle
- H100-SXM: three users own the idle hours. Do they all need fast inter-GPU communication?
- L40S: 12 users own 98% of the idle hours
- The one-session rule and batch-only multi-GPU bound all of this by construction

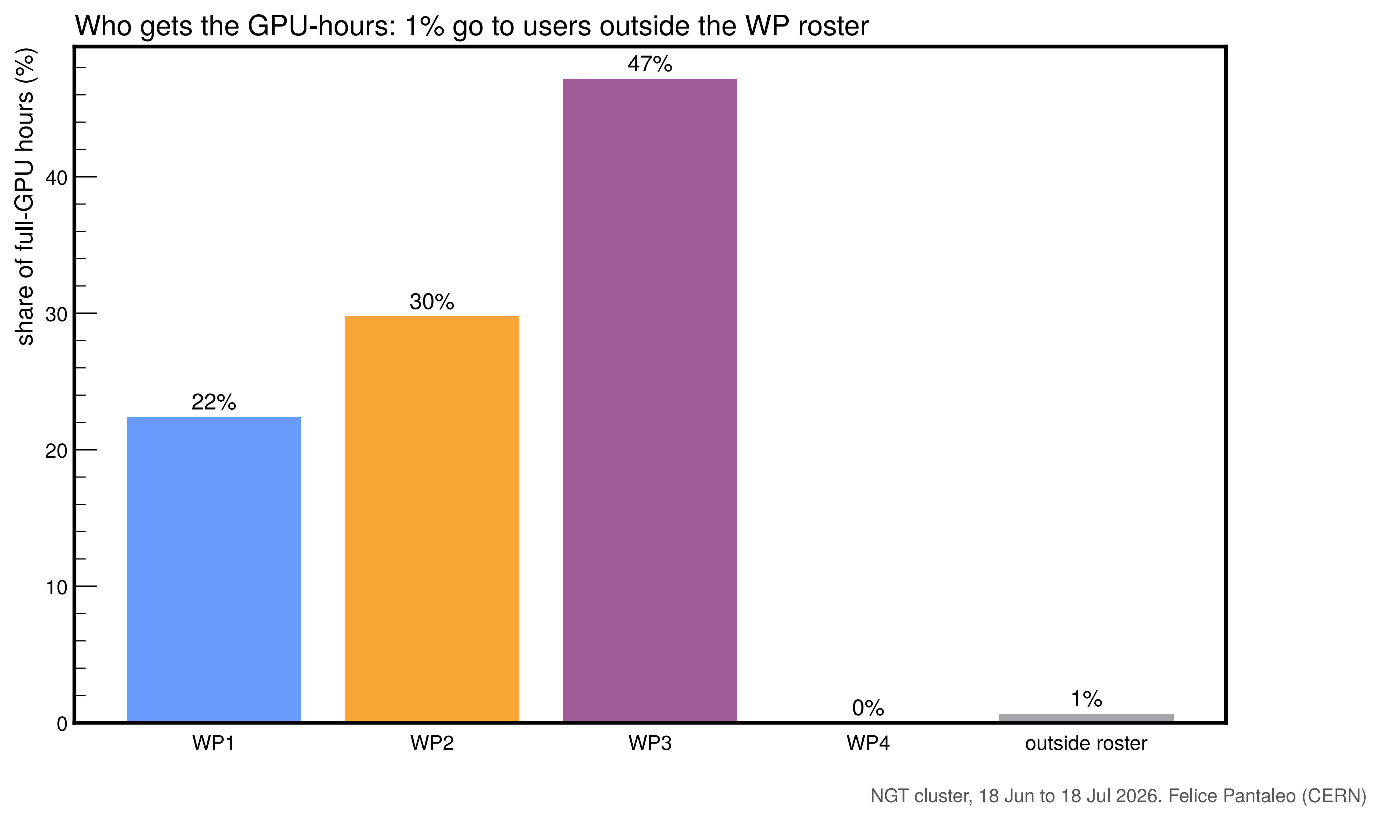


Cloud-equivalent cost



Rental equivalent: 523 kCHF/month, of which at least 289 kCHF idle (3.5 MCHF/year). On-demand rates, July 2026 (AWS).

GPU-hours by WP



Proposal for a new policy

Simulated the same 30 days, same requests, replayed through the proposed policy:

	Today (FCFS)	Proposed
wait p95	965 min	**0 min**
requests satisfied	95.7%	**99.4%**
dev sessions within 15 min	88%	**100%**
NVL idle-held GPU-hours	38 000	**14 300**
running work terminated by the system	none	**none**

> **Policy: one interactive session per member (a new session supersedes the previous one); multi-GPU beyond a 96 GPU-h/month interactive allowance runs as batch behind a WP fair-share priority queue.**
> **Fixed-behavior replay, identical requests for every policy; the FCFS baseline reproduces the observed waits at the median, and to within a factor of about three at p95.**

Proposal to the PMC

One interactive session

One single-GPU session per member, guaranteed and always available; opening a new one replaces the old.

Multi-GPU is batch by default

Submitted, executed, exits at completion. A few hundreds GPU-h/month interactive allowance covers debugging and benchmarking. Amount to be tuned.

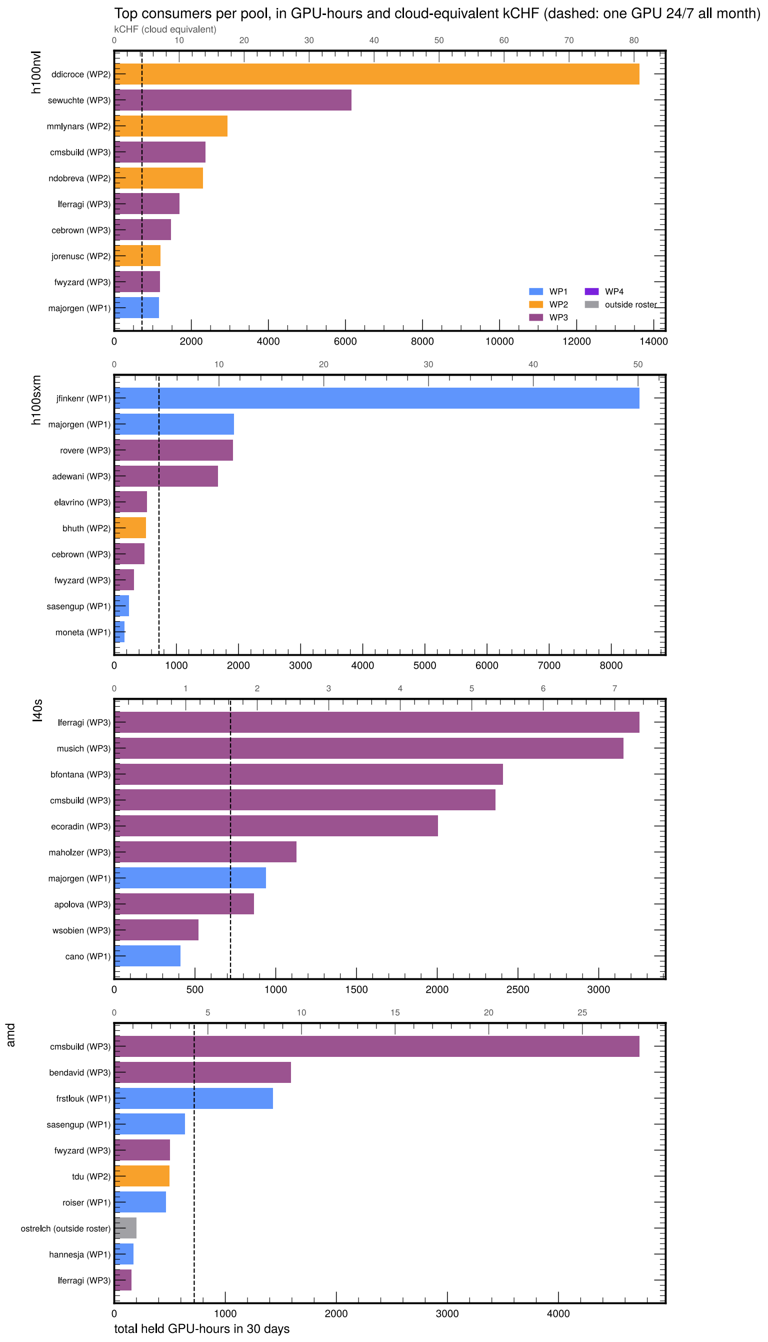
Account and adjust

GPU time charged to WPs with model factors. All parameters simulated and tunable on the real trace.

Backup

Backup: top consumers per pool

- Secondary axis: cloud-equivalent kCHF, cheapest of AWS/GCP, July 2026
- Dashed: one GPU 24/7 all month
- 24 users held more than one GPU-month



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Backup: node cordon timeline

One row per node; label % = share of the month cordoned.

- A few nodes chronically drained: one SXM 43% (1 to 15 Jul), one L40S 73%, MI300X and storage ~100%
- The 6 to 8 July burst of short bars is the coordinated maintenance